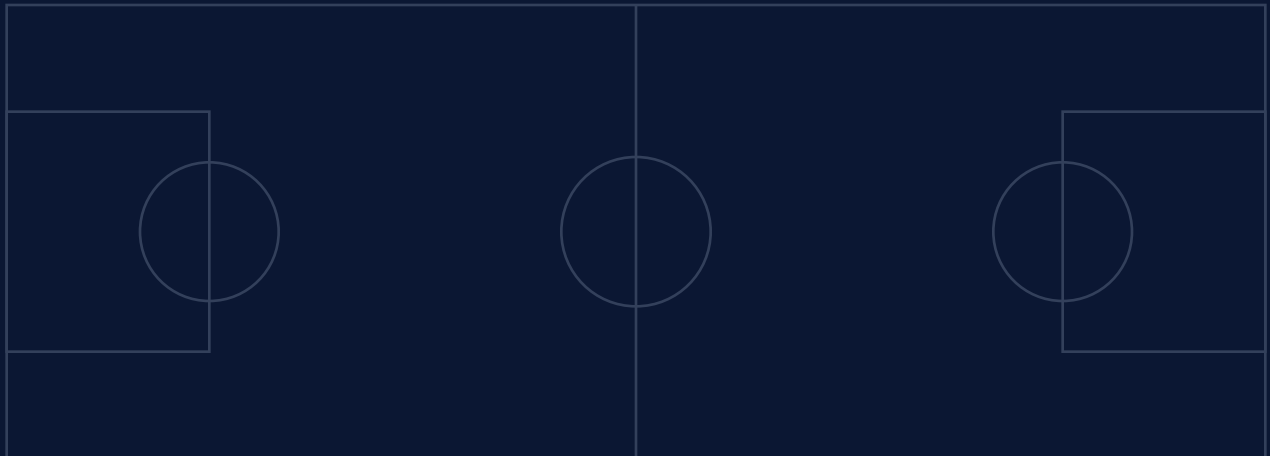


NBA ALL-STAR PREDICTION & SNUB EXPLORATION

A personal machine-learning project about what NBA All-Star selections reward, which players a box-score model overlooks, and what the model's disagreements reveal about basketball data.



INSPIRED BY

Michael Lewis, "The No-Stats All-Star" - Shane Battier

The project began with a simple prediction problem and gradually became an investigation of what conventional statistics make visible - and what they leave out.

The question behind the model

I started with a simple question: can conventional NBA player statistics predict who becomes an All-Star? The idea came from reading Michael Lewis's article about Shane Battier, a player whose value was often described as much larger than his traditional box score suggested.

THE PROBLEM I EXPECTED

Points, rebounds and assists are easy to record, compare and reward. Basketball IQ, defensive positioning, communication and decision quality are much harder to represent. Even a recorded stat can hide very different kinds of value: an assist from creating an open layup and a final pass before a contested make both count as one assist.

I had the same concern with rebounds and steals. They are useful, but simply accumulating more does not guarantee the same increase in positive team impact. That tension made All-Star selection a useful test: the labels are real and easy to obtain, but the reasons behind them are much more complicated.

PREDICT

Learn the statistical profile associated with official All-Stars.

COMPARE

Rank players by probability, not only a yes/no result.

INTERPRET

Use extreme disagreements to ask what the data or selection process may be missing.

How the project evolved

1

2023 PROTOTYPE

I used the most recent complete season available to me and started with conventional per-game statistics. At this point the goal was mainly to see whether I could build a working classifier at all.

2

MORE SEASONS

I expanded the data to 2015-2025. That gave the model more player-seasons while keeping the sample within relatively recent NBA history so older eras would not dominate the patterns.

3

PROBABILITY RANKING

Instead of only asking whether the model predicted All-Star or not, I began looking at probabilities. That made near-misses, extreme snubs and surprising actual selections much easier to study.

4

NASH BENCHMARK

Steve Nash became a test case because his MVP and All-Star success looks unusual if a model focuses too heavily on scoring volume. I began experimenting with assists, efficiency and team winning.

5

OVERFITTING REALIZATION

Eventually I noticed I was changing the model partly because I wanted Nash to receive the answer I already believed was correct. That was a useful lesson: a benchmark can quietly become a target.

6

RICHER-DATA DIRECTION

The project ended with a broader idea. Better basketball evaluation may require different data - tracking, spatial or heat-map information, defensive context and possession-level decisions - rather than only more box-score columns.

How Steve Nash changed the project

Nash became a stress test because his All-Star and two-time MVP recognition is unusual relative to a simple volume-stat profile. He was valuable through playmaking, efficiency, decision-making and team offense, not just through scoring totals.



THE OVERFITTING WARNING

When the early model barely recognized a Nash-style profile, I started giving more importance to assists, efficiency and winning. Those are legitimate parts of his value, but I eventually realized I was also changing the model until a player with a known answer received that answer. Nash was becoming a target instead of a benchmark.

That changed how I treated model errors. A disagreement did not automatically mean I needed another feature or a different weight. It could mean the model was missing context, the label itself depended on context, or the available statistics were simply not capable of representing the type of value I cared about.

ERA EFFECT

The same raw number means something different when pace, scoring, shot selection, positions and league style change. Using 2015-2025 reduced the problem, but did not remove it.

FUTURE QUESTION

Instead of assuming one timeless standard, a future model could ask whether the same player profile would look All-Star caliber in different NBA eras.

This was the point where the project became less about maximizing one accuracy score and more about understanding what the model was actually learning.

Russell Westbrook - 2021

WAS | PG | Age 32

NOT SELECTED

98.8%

MODEL ALL-STAR PROBABILITY



The model saw an obvious All-Star statistical profile.

PTS 22.2	AST 11.7	REB 11.5	STL 1.4	BLK 0.4	TOV 4.8
FG% 43.9%	3P% 31.5%	eFG% 47.4%	FT% 65.6%	G 65	MPG 36.4

WHY THE MODEL PRODUCED THIS RESULT

A 22.2-point triple-double season gave the model exactly the kind of production it rewarded. Scoring, assists and rebounding were all unusually high, especially for a guard, so several of the strongest model signals pointed in the same direction.

WHY THIS CASE MATTERS

This is also a warning about counting statistics. A rebound or assist enters the data as a positive event, but the model does not know the quality of the possession, the efficiency cost, defensive tradeoffs or whether those totals translated into the same level of team impact.

HOW I READ THE RESULT

I kept this case because 98.8% is such a strong disagreement. It made the model easier to understand: it was recognizing a statistical shape, not making a complete basketball judgment.

The 98.8% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that Russell Westbrook deserved or did not deserve a selection.

Domantas Sabonis - 2024

SAC | C | Age 27

NOT SELECTED

94.0%

MODEL ALL-STAR PROBABILITY



One of the clearest predicted snubs in the saved results.

PTS 19.4	AST 8.2	REB 13.7	STL 0.9	BLK 0.6	TOV 3.3
FG% 59.4%	3P% 37.9%	eFG% 60.9%	FT% 70.4%	G 82	MPG 35.7

WHY THE MODEL PRODUCED THIS RESULT

Sabonis produced across three major box-score categories at once: 19.4 points, 13.7 rebounds and 8.2 assists. That combination is rare, and his strong shooting efficiency made the profile look even more like the selected All-Stars in the training data.

WHY THIS CASE MATTERS

The model does not see conference competition, roster limits, voting, narrative or how the field looked at the exact time selections were made. A high probability therefore says his season resembled learned All-Star profiles, not that the official decision was objectively wrong.

HOW I READ THE RESULT

This is one of the best examples of why I moved from a simple yes/no classifier toward probability ranking and 'snub' exploration.

The 94.0% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that Domantas Sabonis deserved or did not deserve a selection.

Anthony Davis - 2023

LAL | C | Age 29

NOT SELECTED

90.8%

MODEL ALL-STAR PROBABILITY



Elite scoring, rebounding and rim protection made him strongly All-Star-like.

PTS 25.9	AST 2.6	REB 12.5	STL 1.1	BLK 2.0	TOV 2.2
FG% 56.3%	3P% 25.7%	eFG% 57.3%	FT% 78.4%	G 56	MPG 34.0

WHY THE MODEL PRODUCED THIS RESULT

Davis had several obvious positive signals at the same time: 25.9 points, 12.5 rebounds, 2.0 blocks and efficient two-point scoring. A model built around season averages naturally reacts strongly to that combination.

WHY THIS CASE MATTERS

Season averages hide timing. They do not explain when games were missed, availability around voting, replacement rules or the exact context voters had when choosing the roster.

HOW I READ THE RESULT

This case helped separate 'great season statistics' from 'the real selection process.' Those are related targets, but they are not identical.

The 90.8% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that Anthony Davis deserved or did not deserve a selection.

James Harden - 2023

PHI | PG | Age 33

NOT SELECTED

85.4%

MODEL ALL-STAR PROBABILITY



Scoring and elite playmaking created a strong statistical case.

PTS 21.0	AST 10.7	REB 6.1	STL 1.2	BLK 0.5	TOV 3.4
FG% 44.1%	3P% 38.5%	eFG% 53.6%	FT% 86.7%	G 58	MPG 36.8

WHY THE MODEL PRODUCED THIS RESULT

Harden gave the model two major signals it could measure easily: 21.0 points and 10.7 assists per game. His shooting percentages were also respectable, so the probability was not coming only from inefficient volume.

WHY THIS CASE MATTERS

Human selection can depend on role, availability, competition and expectations for a player. The model has none of that context unless those variables are explicitly supplied.

HOW I READ THE RESULT

This case reinforced that reproducing a human award process is different from building a general measure of player value.

The 85.4% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that James Harden deserved or did not deserve a selection.

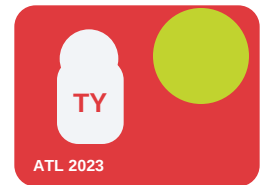
Trae Young - 2023

ATL | PG | Age 24

NOT SELECTED

85.1%

MODEL ALL-STAR PROBABILITY



High-volume offensive creation fit the learned profile.

PTS 26.2	AST 10.2	REB 3.0	STL 1.1	BLK 0.1	TOV 4.1
FG% 42.9%	3P% 33.5%	eFG% 48.5%	FT% 88.6%	G 73	MPG 34.8

WHY THE MODEL PRODUCED THIS RESULT

Young's 26.2 points and 10.2 assists are exactly the kind of visible production a basic model can recognize. Both are historically associated with star offensive roles, so they pushed his probability upward.

WHY THIS CASE MATTERS

The same case returns to the original Battier-inspired problem: offense is much easier to count than defensive positioning, off-ball decisions and many possession-level effects. The model can overstate what is easy to measure.

HOW I READ THE RESULT

Young is useful because his result shows how a model can be internally consistent while still being incomplete as a basketball evaluator.

The 85.1% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that Trae Young deserved or did not deserve a selection.

Jimmy Butler - 2023

MIA | PF | Age 33

NOT SELECTED

48.8%

MODEL ALL-STAR PROBABILITY



The most useful borderline case.

PTS 22.9	AST 5.3	REB 5.9	STL 1.8	BLK 0.3	TOV 1.6
FG% 53.9%	3P% 35.0%	eFG% 56.0%	FT% 85.0%	G 64	MPG 33.4

WHY THE MODEL PRODUCED THIS RESULT

Butler had strong efficiency, scoring and steals, but he did not have the extreme counting-stat volume of several players above him. That left him almost exactly on the model's boundary.

WHY THIS CASE MATTERS

A 48.8% result is more informative than forcing a hard 'no.' It shows uncertainty and makes it easier to compare borderline profiles rather than pretending every classification has equal confidence.

HOW I READ THE RESULT

This was one reason probability ranking became more interesting to me than accuracy alone.

The 48.8% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that Jimmy Butler deserved or did not deserve a selection.

Andrew Wiggins - 2022

GSW | SF | Age 26

ACTUAL ALL-STAR

2.4%

MODEL ALL-STAR PROBABILITY



The model strongly rejected an actual selection.

PTS 17.2	AST 2.2	REB 4.5	STL 1.0	BLK 0.7	TOV 1.5
FG% 46.6%	3P% 39.3%	eFG% 54.3%	FT% 63.4%	G 73	MPG 31.9

WHY THE MODEL PRODUCED THIS RESULT

Wiggins' 17.2 points, 4.5 rebounds and 2.2 assists were modest compared with the high-volume statistical profiles the model learned to reward. His efficient three-point shooting helped, but not enough to overcome the lower totals.

WHY THIS CASE MATTERS

Actual All-Star selection is not one universal formula. Team success, role, voting, conference context and other factors can matter. Those can produce a legitimate selection that looks unusual to a box-score model.

HOW I READ THE RESULT

This is a useful false negative because it shows the danger of assuming that official labels can always be reconstructed from a small set of player averages.

The 2.4% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that Andrew Wiggins deserved or did not deserve a selection.

Draymond Green - 2022

GSW | PF | Age 31

ACTUAL ALL-STAR

3.6%

MODEL ALL-STAR PROBABILITY



The case closest to the Shane Battier inspiration.

PTS 7.5	AST 7.0	REB 7.3	STL 1.3	BLK 1.1	TOV 3.0
FG% 52.5%	3P% 29.6%	eFG% 55.6%	FT% 65.9%	G 46	MPG 28.9

WHY THE MODEL PRODUCED THIS RESULT

Green's value was not built around scoring volume. Although his assists, rebounds, steals and blocks show versatility, the original feature set still gave the model far fewer loud signals than it gave high-scoring stars.

WHY THIS CASE MATTERS

Defense, communication, screening, positioning, rotations and lineup effects are difficult to summarize with basic box scores. A low model probability can therefore reveal a limitation of the representation rather than a limitation of the player.

HOW I READ THE RESULT

This result connects most directly to the question that started the project: which valuable players become statistically quiet when the available data emphasize the wrong things?

The 3.6% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that Draymond Green deserved or did not deserve a selection.

Al Horford - 2018

BOS | C | Age 31

ACTUAL ALL-STAR

5.5%

MODEL ALL-STAR PROBABILITY



A balanced two-way profile the model undervalued.

PTS 12.9	AST 4.7	REB 7.4	STL 0.6	BLK 1.1	TOV 1.8
FG% 48.9%	3P% 42.9%	eFG% 55.3%	FT% 78.3%	G 72	MPG 31.6

WHY THE MODEL PRODUCED THIS RESULT

Horford's scoring and rebounding were not extreme, even though his season also included passing, efficient shooting, defense and an important role on a strong team. The model's simple inputs made that balanced contribution look less star-like.

WHY THIS CASE MATTERS

This is another example of value being spread across several areas instead of concentrated in one huge statistic. Models can miss that when a few volume features dominate.

HOW I READ THE RESULT

I kept Horford because his profile sits between Green's hard-to-measure impact and the louder traditional production of most All-Stars.

The 5.5% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that Al Horford deserved or did not deserve a selection.

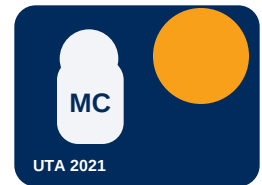
Mike Conley - 2021

UTA | PG | Age 33

ACTUAL ALL-STAR

5.8%

MODEL ALL-STAR PROBABILITY



Another selected player without a loud box-score profile.

PTS 16.2	AST 6.0	REB 3.5	STL 1.4	BLK 0.2	TOV 1.9
FG% 44.4%	3P% 41.2%	eFG% 55.2%	FT% 85.2%	G 51	MPG 29.4

WHY THE MODEL PRODUCED THIS RESULT

Conley's 16.2 points and 6.0 assists were good, but they were not extreme compared with the profiles receiving the highest probabilities. His shooting efficiency and low turnovers add context that a volume-heavy model may not reward enough.

WHY THIS CASE MATTERS

Role, team context, efficiency and the circumstances of a selection can matter even when the season line does not look like a stereotypical star line.

HOW I READ THE RESULT

Cases like Conley changed how I interpreted errors. Instead of immediately trying to 'fix' every disagreement, I started asking what information the model never received.

The 5.8% probability should be read as similarity to patterns learned from official All-Stars, not as an objective percentage that Mike Conley deserved or did not deserve a selection.

What the project taught me

01 Counting statistics are powerful but incomplete.

They are easy for a model to learn, which can make measurable production dominate harder-to-measure impact.

02 Selection and value are different targets.

Predicting who becomes an All-Star is not identical to measuring who is most valuable.

03 Disagreement is useful.

False positives and false negatives can reveal missing context instead of being treated only as errors.

04 Data type matters as much as model choice.

Tracking, heat maps, defensive context and possession-level information may add more understanding than repeatedly changing weights on the same basic statistics.

05 Era should be modeled.

A future version could compare a player's profile with era-specific standards instead of assuming one timeless definition of an All-Star.

06 The process was part of the project.

I began with very little Python experience. Building, revising and breaking versions taught me pandas, scikit-learn, data preparation, model evaluation and how easy it is to overfit an idea.

FINAL PERSPECTIVE

The project started with: 'Can statistics predict an All-Star?' It ended with a better question: 'What does the available data make visible, and what does it hide?'

Technical work: Python / Jupyter / pandas / scikit-learn / Random Forest / logistic regression / feature engineering
probability ranking / model evaluation / data cleaning / exploratory visualization